Q32 — Deeper projects

Quantifying information flow with entropy

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1. **Entropy-triggered abstention.** Build a selective-prediction wrapper around a small instruction model on a trivia or multiple-choice set. Use predictive entropy (and the logit margin) as the abstention score; plot the accuracy–coverage (risk–coverage) curve and compare entropy against a verbalised "how sure are you?" baseline. *Deliverable:* the curve + a short note on where entropy fails as a confidence proxy. (Links to Q10–Q12.)

2. **Reproduce the binning artefact, then kill it.** Take a 2–3 layer MLP on a toy classification task. Reproduce a "compression-phase" information-plane plot with `tanh` + coarse binning, then show it weaken/vanish with (a) `ReLU`, (b) a finer/adaptive binning, and (c) a KSG estimator. *Deliverable:* three plots and a paragraph on what, if anything, survives.

3. **Attention-entropy atlas.** For a small open transformer, compute per-head attention entropy across layers on a fixed prompt set; cluster heads by entropy profile and cross-reference against known head taxonomies (induction heads, attention sinks, positional heads). *Deliverable:* a heatmap + a claim about which clusters are real vs. which are averaging artefacts, tested by ablation.

4. **Estimate vs. intervene bake-off.** Pick one specific information-flow hypothesis (e.g. "the subject token's information reaches the final position through layer ℓ"). Answer it two ways — a variational MI bound (InfoNCE) *and* activation patching — and write up where they agree, where they diverge, and which you would trust in a paper.

5. **LLM-as-compressor.** Reimplement the Delétang et al. experiment at toy scale: use a frozen small LM as the probability model inside an arithmetic coder and measure bits/token on held-out text; compare against `gzip` and against the LM's reported perplexity. *Deliverable:* a table showing compression ratio = perplexity in different units.